MIL-C-24643/44B
3 October 1986
SUPERSEDING
MIL-C-0024643/44A(SH)
8 November 1985
MIL-C-24643/44
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPE LS3SF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, AWG no. 18, class B stranding (7 by 0.0152 inch), uncoated. Maximum diameter 0.050 inch.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation. Nominal thickness 0.020 inch. Colored insulation - one conductor black, one conductor white and one conductor red for each triad.
Fourth - One black, one white and one red conductor cabled together with a nominal lay of 1-1/2 inches to form a triad. Fillers as necessary in order to form a round assembly.
Fifth - One polyester tape, type G in accordance with MIL-I-631, applied helically with overlap.
Sixth - Braided shield of AWG no. 36 uncoated or tin-coated copper. Braid angle 30 to 35 degrees; minimum coverage of 85 percent.
Seventh - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, sealed. Standard identification code applied by method 2 on inner tape.
- Diameter over shield insulation 0.240-inch nominal.

(B) denotes changes.

AMSC N/A FSC 6145
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Eighth - The specified number of triads (see table I) cabled together with a lay not greater than 16 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with triad no. 1, from center outward. A single triad shall not be used as the central core of the cable. Fillers shall be employed as necessary to form a firm, round assembly.

Ninth - Binder tape of polyester, type G in accordance with MIL-I-631, applied helically with overlap.

Tenth - Cross-linked polyolefin jacket double layer, reinforced. Minimum thickness 0.100 inch. Cable surface marking required.

TABLE I. Details.

Military part no. M24643/44	Type and size	Number of triads	Overall diameter	
			Minimum (inches)	Maximum (inches)
-01UN	LS3SF-7	7	1.000	1.040

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	7.06
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor, within triads.....	2000
Conductor to shield, within triads.....	1000
Shield to shield, between triads.....	500
Insulation resistance - megohms/1000 feet, minimum.....	500
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Mutual capacitance - within triads, at 1 kHz, pF/ft, maximum.....	30
Capacitance unbalance - percent maximum.....	5
Group B:	
Cold bending, cable - at minus 30 ± 2°C, maximum mandrel diameter 12 times cable diameter.....	No damage
Bending endurance - at 25 ± 2°C, cycles, minimum.....	2000
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum.....	1800
Elongation - percent, minimum.....	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum.....	1300
Elongation - percent, minimum.....	160

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Permanence of printing (jacket) - cycles, minimum	250
Heat distortion - percent of unaged, maximum	30
Cable filler removability	No failure
Shield - conformance to material, construction and coverage	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Aging and compatibility (cable) (125 ± 5°C)	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	--
(B) Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

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UNIT ORDERING LENGTH: 1000 feet (nominal)

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-44)

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